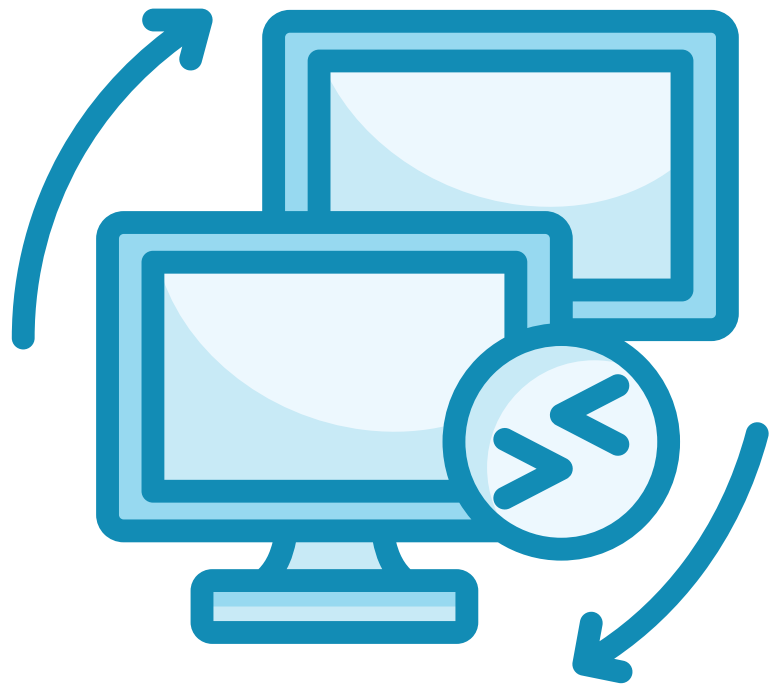




AZURE VIRTUAL DESKTOP

AZURE VIRTUAL DESKTOP



- This is a desktop and app virtualization service that runs on Azure.
- Here you can offer virtual desktops to users. These desktop machines can run Windows 11, Windows 10 or Windows Server.
- You can offer this to a single user or use multi-session capabilities.



Describe Azure architecture and
services - Azure compute

Revision



Azure compute

1

Using a cloud platform like Azure comes with a ton of benefits.

2

You don't need to buy or deploy physical servers. This allows you to save on the initial set of costs.

3

In an on - premises setup, if you had to extend your architecture, you would need to buy more servers and storage.

4

But on Azure, you can extend your compute and storage at any point in time. Gives you that extra level of flexibility.

Azure compute

Azure virtual machines

Here the underlying physical infrastructure is managed for you. You can create and delete machines whenever required.

Azure Web Apps

This provides a managed service for your web applications. You can deploy and scale your web apps accordingly.

Azure Functions

This is a serverless service. You can just deploy your code and run on demand. You only get charged based on the usage.

Azure Container Instances

This service allows you to host your container - based applications. It manages the underlying infrastructure for you.

Azure compute

1

Virtual Machine scale sets allow you to deploy an identical set of machines to host your application.

2

You can define rules that can increase the number of machines in the scale set automatically.

3

You can also defined rules that can decrease the number of machines in the scale set automatically.

4

When there is a heavy load, increase the number of machines.
When the load decreases , decrease the number of machines.

Azure compute

1

If you are deploying an application on multiple virtual machines, and you need to protect the application against data center level failures, deploy the machines to multiple Availability zones.

2

Most of the Azure regions have support for Availability zones. Check the Microsoft documentation to confirm the availability.

3

You can deploy machines which can be either Windows or Linux to Availability zones.

Azure compute

1

Using Azure virtual desktop, you can give virtual desktop access to your users.

2

There is support for Windows 10, Windows 11, Windows Server.

3

There is support for both desktop and application virtualization.

Azure compute

Your team needs to make use of a serverless compute service on Azure.
Which of the following can be used for this requirement?

Azure virtual machines

Azure virtual machine scale set

Azure Functions



Azure compute

You need to setup a workload on a set of Azure virtual machines. You want to take advantage of the unused capacity that Azure has to offer so that you can save on costs on compute for the machines. Which of the following could you make use of for this requirement?



Azure Spot virtual machines

Azure on-demand pricing

Azure reserved pricing

Azure compute

You have to deploy an application on a set of Azure virtual machines. You have to ensure that the application is available even if a data center becomes unavailable within the region. Which of the following would you implement for this requirement?

Deploy machines as part of an Availability set.

Deploy machines as part of a scale set.



Deploy machines as part of many Availability zones.

Azure compute

Which of the following can be used to run application code without the need of having a server in place for hosting the code.

Azure virtual machines

Azure Functions



Azure Virtual Machine Scale sets

Azure compute

You need to deploy a container - based application onto Azure. Which of the following is a managed service that can be used for this requirement?



Azure container instance

Azure Functions

Azure container registry

Azure compute

You're planning on getting reservation pricing when it comes to the Azure virtual machine service. Which of the following are valid durations when it comes Azure reservations?



One year

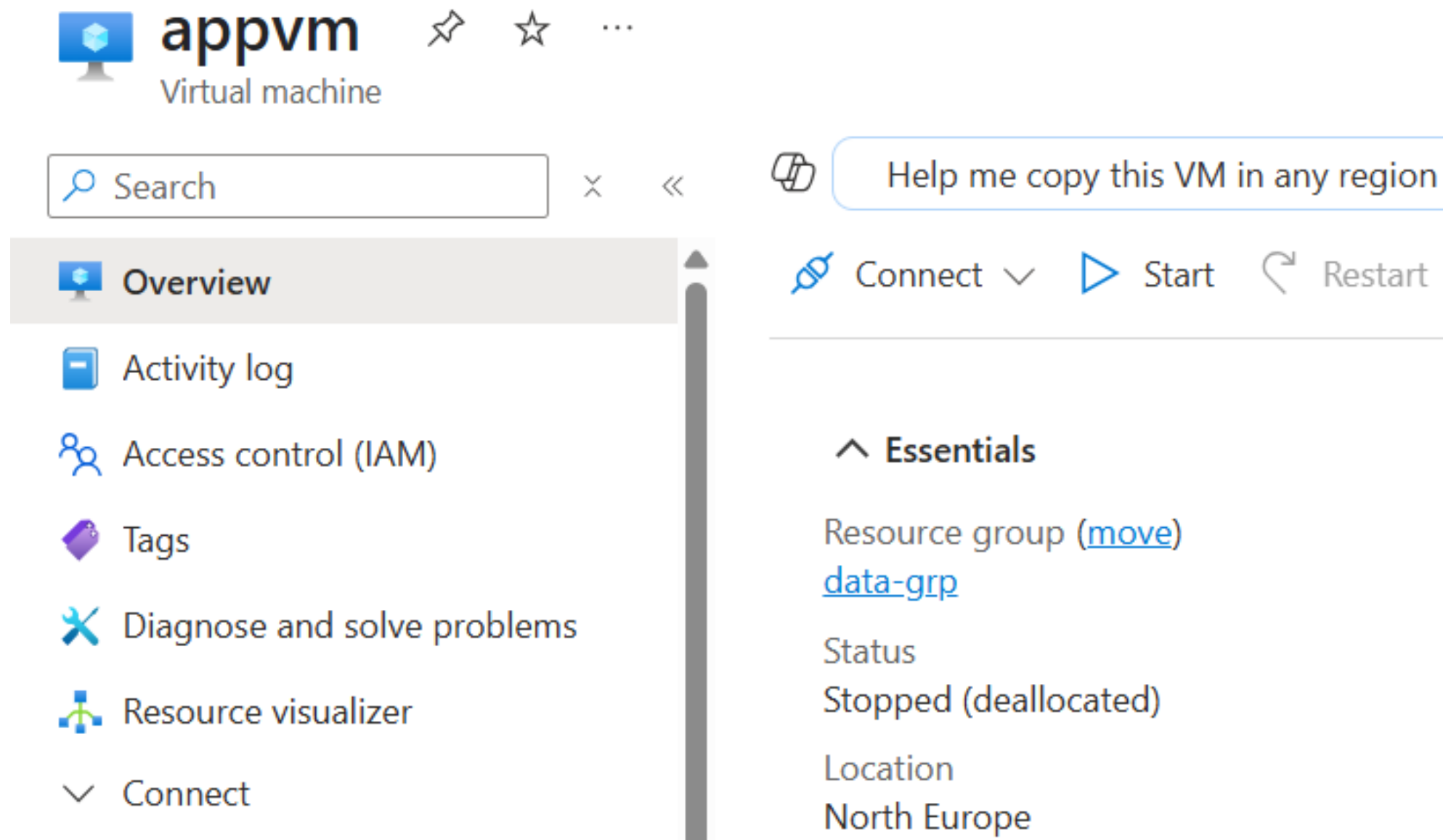
Two years



Three years

Azure compute

You have an Azure virtual machine in the following state. What will you be charged for when it comes to the virtual machine setup?



The screenshot shows the Azure portal interface for a virtual machine named 'appvm'. The left sidebar contains navigation links: Overview (selected), Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, and Connect. The main content area displays the VM's status as 'Stopped (deallocated)' and its location as 'North Europe'. A red checkmark is placed next to the 'Disks' section in the diagram on the right.

appvm
Virtual machine

Search

Help me copy this VM in any region

Connect Start Restart

Essentials

Resource group ([move](#))
[data-grp](#)

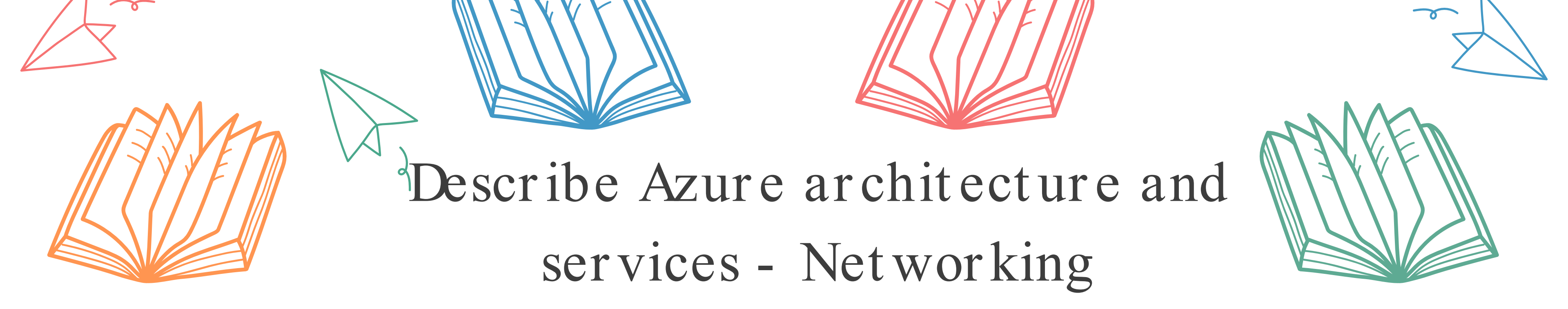
Status
Stopped (deallocated)

Location
North Europe

Compute

Disks

Virtual Network



Describe Azure architecture and
services - Networking

Revision



Azure networking

Azure virtual network

This is an isolated network on Azure cloud.

Virtual network peering

This can be used to connect virtual networks together.

ExpressRoute

This provides a dedicated connection between on - premises , Azure and other Microsoft services. Here the traffic does not flow via the Internet.

VPN connections

This is a secure and encrypted connection that can be setup between Azure and clients.

Azure networking

1

Azure network security groups can be used to filter network traffic between Azure resources in an Azure virtual network.

2

Rules can be defined to allow or deny inbound and outbound traffic.

3

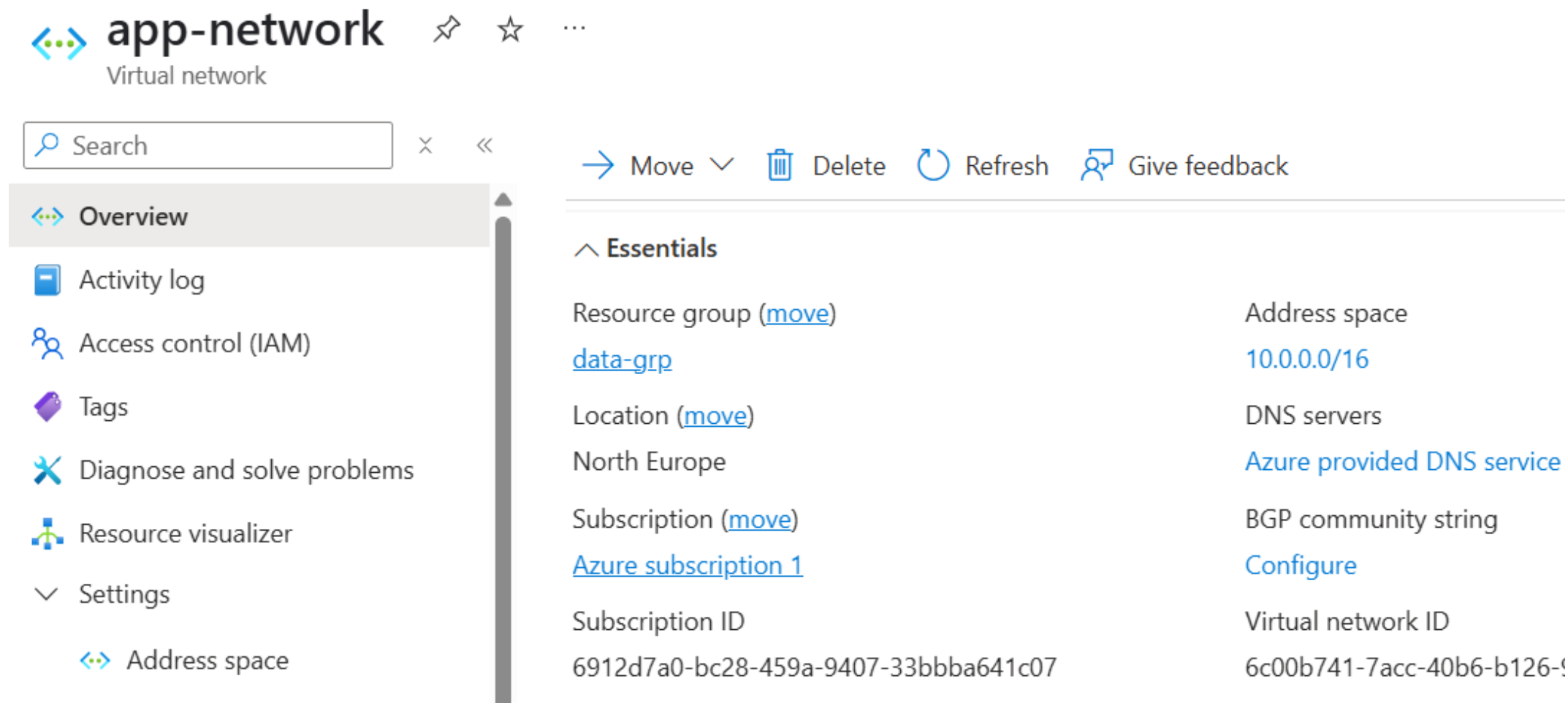
There are already default rules in place. But you can add your own rules as well.

4

A network security group can be attached to a subnet or to a network interface.

Azure networking

You have a virtual network as shown below.



The screenshot shows the Azure portal interface for a virtual network named 'app-network'. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, and Address space. The main content area displays the 'Overview' page for the virtual network. At the top, there are action buttons: Move, Delete, Refresh, and Give feedback. Below this, the 'Essentials' section lists key properties:

Property	Value
Resource group	data-grp
Location	North Europe
Subscription	Azure subscription 1
Subscription ID	6912d7a0-bc28-459a-9407-33bbba641c07
Address space	10.0.0.0/16
DNS servers	Azure provided DNS service
BGP community string	Configure
Virtual network ID	6c00b741-7acc-40b6-b126-!



Yes

No

Can you create another virtual network with the name of app - network and address space of 10.0.0.0/16 in another resource group.

Azure networking

Which protocol is used by Azure ExpressRoute



Border Gateway Protocol

Internet Gateway Protocol

Routing Information Protocol

Azure networking

When deploying an Azure virtual machine , which of the following is a mandatory resource that must be present for the virtual machine?

Public IP address

Network Security Group



Virtual network

Azure networking

When using Azure ExpressRoute to provide connectivity between on premises and the Microsoft Cloud, at which layer of the OSI model does ExpressRoute provide the connectivity?

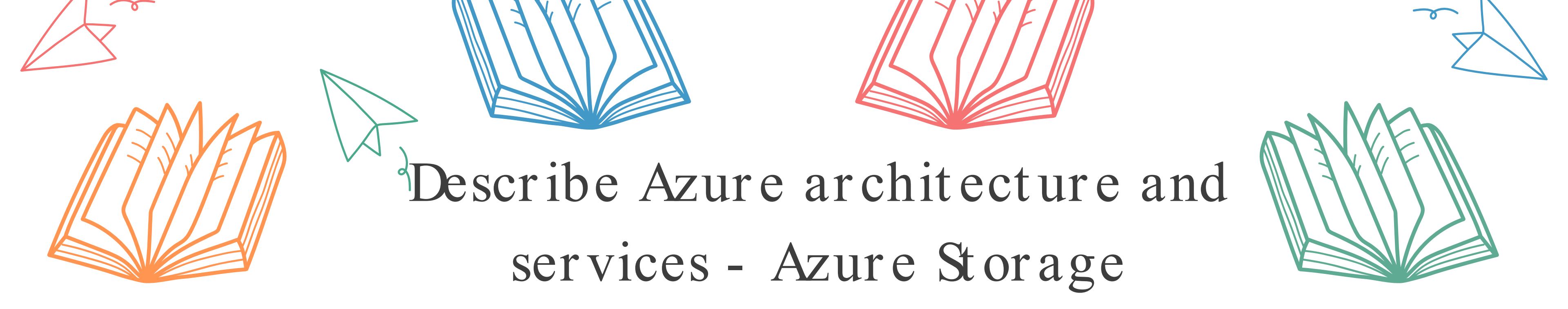
-

Layer 2

Layer 3



Layer 7



Describe Azure architecture and
services - Azure Storage

Revision



Azure storage

Azure Blob

This is object level storage. Each object gets its own unique URL.

Files

This allows you to host file shares. You can map to these file shares from client machines.

Queue

This allows you to provide a messaging layer between applications or application components.

Table

This service allows you to store non-relational structured data.

Azure storage

Hot tier

Optimized for data that is accessed or modified frequently. High storage and low access costs.

Cool tier

Optimized for data that is infrequently accessed or modified. The data needs to be stored for a minimum of 30 days.

Cold tier

Optimized for data that rarely accessed or modified. But we need quick retrieval of data. Here the data needs to be stored for a minimum of 90 days.

Archive tier

Optimized for data that is rarely accessed. Can take time to access the data. Data should be stored for a minimum of 180 days.

Azure storage

You need to migrate large amounts of data (TB's of data) from your on premises data center onto Azure. Which of the following can be used to migrate the data in a reliable way and least expensive way.

Use Azure Storage Explorer

Make use of Azure Data Box



Use Azure ExpressRoute

Azure storage

Which of the following can be used to copy blobs from a local file system onto the Blob service of an Azure Storage account.



Azure Storage Explorer

AzCopy



Azure File Sync

Azure storage

You have the following Azure storage account.

The screenshot shows the Azure portal interface for a storage account named 'appstore5554554'. The left sidebar contains navigation links: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, and Partner solutions. The main content area is divided into two sections. The 'Essentials' section on the left lists: Resource group (data-grp), Location (northeurope), Subscription (Azure subscription 1), Subscription ID (6912d7a0-bc28-459a-9407-33bbba641c07), and Disk state (Available). The 'Properties' section on the right lists: Performance Standard, Replication Locally-redundant storage, Account kind StorageV2 (general purpose), Provisioning state Succeeded, and Created 5/27/2025, 12:14:12. A red checkmark is placed next to the 'Performance Standard' and 'Replication Locally-redundant storage' properties.

Resource group (move)	Performance Standard
Location northeurope	Replication Locally-redundant storage
Subscription (move) Azure subscription 1	Account kind StorageV2 (general purpose)
Subscription ID 6912d7a0-bc28-459a-9407-33bbba641c07	Provisioning state Succeeded
Disk state Available	Created 5/27/2025, 12:14:12

2

3

4

How many copies of your data are maintained by the storage account?

Azure storage

You have the following Azure storage account.

Yes

No

The screenshot shows the Azure portal interface for a storage account named 'fileshareaccount5000'. The left sidebar contains navigation links: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, and Storage browser. The main content area shows the 'Overview' page with a search bar and a toolbar with icons for Upload, Open in Explorer, Delete, Move, Refresh, Open in mobile, and Feedback. Below the toolbar, the 'Essentials' section displays a table of account details:

Resource group (...)	: data-grp	Performance	: Premium
Location	: northeurope	Replication	: Locally-redundant storage (LRS)
Subscription (move)	: Azure subscription 1	Account kind	: FileStorage
Subscription ID	: 6912d7a0-bc28-459a-9407-33bbba641c07	Provisioning state	: Succeeded
Disk state	: Available	Created	: 5/27/2025, 6:57:04 PM
Tags (edit)	: Add tags		

A red checkmark is placed next to the 'Performance' field, which is set to 'Premium'.

Can you set the access tier for files within the file shares in the storage account?

The header features a white background with several hand-drawn illustrations. There are four open books in orange, blue, red, and green. Additionally, there are four paper airplanes in the same color palette, some with small circular motion lines indicating flight. The text "Describe Cloud Concepts" is centered in a black serif font.

Describe Cloud Concepts

Revision



Cloud Concepts

1

Infrastructure as a service - Here Azure maintains the underlying physical hardware.

2

You maintain the operating system, the applications and the data.

3

This gives you control over various aspects such as installing and configuring the application.

4

An example is the usage of Azure virtual machines.

Cloud Concepts

1

Platform as a service - Here Azure maintains the underlying physical hardware, the operating system and the runtime.

2

You maintain the the application and the data.

3

Here the company does not want the burden of managing the virtual infrastructure.

4

An example is the usage of Azure SQL databases. Here you can directly start working with the database system.

Cloud Concepts

1

Software as a service - Azure maintains the underlying physical hardware, the operating system, the runtime and application.

2

Here you directly start interacting with the application and adding data.

3

An example is Microsoft 365.

Cloud Concepts

1

Using Cloud services helps to reduce both CapEx and Opex costs.

2

Here the company does not need to invest and maintaining data centers, physical servers etc.

3

Also cloud computing platforms provide a pay - as - you - go model where you pay based on how much you use.

4

As cloud platforms grow and start to get more customers, they can start offering more discounts and less service pricing.

Cloud Concepts

Fault tolerance

How tolerant is your system to faults. How can you add measures to make your application system recover from faults.

High Availability

What measures you take to make your overall system highly available. Using Availability sets and zones at the infrastructure level.

Scalability

Demand on your application is increasing. How can you scale to meet the demand. Using Azure Virtual Machine Scale sets.

Disaster Recovery

How can you recover from a service failure or an entire region failure. Replicating application data across regions.

Cloud Concepts

Your team is planning on making use of Microsoft 365 suite of products.
This comes under which cloud service model?

Infrastructure as a service

Platform as a service

Software as a service



Cloud Concepts

Your team is planning on hosting an application on Azure. The plan is to host the machine using the Azure virtual machine scale set service. Which of the following can you accomplish via the use of this service?

Disaster Recovery

Elasticity



Replication

Cloud Concepts

Your team is planning on hosting an application on Azure. The plan is to use the Azure SQL database service. This service comes with which cloud service model?

Infrastructure as a service

Platform as a service



Software as a service

Using AI services



AI services

Azure has a host of in-built AI services. You can use this services to incorporate AI features within your application.



A user can upload a set of images to an application. The application should then generate different tags or divide them into groups based on the objects it finds within the images.



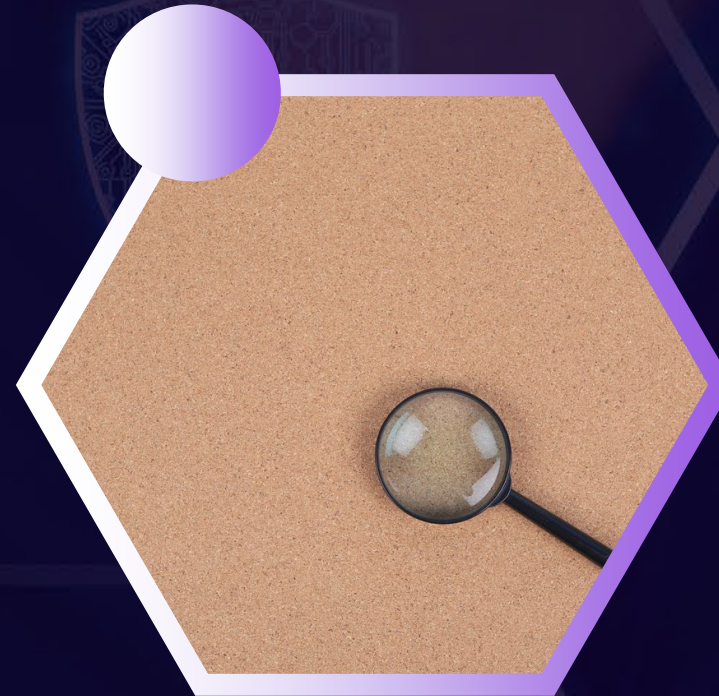
Now instead of developing the functionality of being able to detect objects within an image from scratch , we can leverage existing solutions on Azure to do this for us.

Azure AI Vision



Generate captions

The service can be used to generate captions and descriptions from images.



Detect Objects

We can also detect individual objects from images.



Optical character recognition

We can also extract text from images.

The header features a white background with several hand-drawn illustrations. There are four open books in orange, blue, red, and green. Additionally, there are four paper airplanes in the same color scheme, some with small circular motion lines. The text is centered in a black serif font.

Describe Azure architecture and
services - Identity and Access

Revision



Identity and Access

1

Authentication - This is the process wherein the user's credentials are verified - Example - User ID and Password.

2

Microsoft Entra ID is the identity provider that can perform the act of authentication.

3

Authorization - Here a check is conducted to see if an identity has the required permissions to access a resource.

4

Microsoft Entra ID also performs the act of authorization.

Identity and Access

1

Multi - Factor Authentication - This can be used as an extra security measure during the user login process.

2

Here in addition to entering a username and password, a user can use another mechanism to complete the login process.

3

For example, the user can receive a notification or code on the authenticator app on their mobile device.

4

You can enable Multi - Factor authentication on a per - user basis or use Conditional Access policies.

Identity and Access

1

Conditional Access - This can be used as an extra security measure to access a resource.

2

Here additional signals are used to check if a user can be granted access to a resource.

3

These signals can include to check if a user belongs to a particular group or if the user is logging in from a known location.

4

If the user fulfils the set conditions then access can be granted to the resource.

Identity and Access

1

Microsoft Defender for Cloud - This is a cloud - native application protection platform.

2

You can use this service to protect cloud - based applications from threats and vulnerabilities.

3

To get access to all security benefits , you need to use the paid plans in Microsoft Defender for Cloud.

4

You can protect workloads that run on Azure, Amazon Web Services , Google Cloud Platform and on - premises as well.

Identity and Access

Which of the following in Azure can be used to authenticate users and applications onto resources in Azure?

Microsoft Defender for Cloud

Microsoft Entra ID



Microsoft Sentinel

Identity and Access

Which of the following can be used to check if the Azure account is meeting various compliance and regulatory standards?



Microsoft Defender for Cloud

Microsoft Entra ID

Conditional Access Policies

Identity and Access

Your team is planning on using Microsoft Entra ID. Can you use Microsoft Entra ID as an authentication tool for both Azure and Microsoft 365?



Yes

No

Identity and Access

Which of the following is the term used for
“Making sure the user performs a set of steps during the process of
identification”



Multi - Factor Authentication

Authorization

Resource Provisioning



Describe Azure management and
governance

Revision



Azure management and governance

1

An organization has the ability to create multiple subscriptions as part of an Azure account.

2

As an example, you can create individual subscriptions for each department in the organization.

3

You can also use resource tags to add metadata or additional key value pairs to your resources.

4

Management groups can be used to manage multiple subscriptions.

Azure management and governance

1

We can build and manage Azure resources via the Azure portal.

2

We can build and manage resources via other tools as well - Azure CLI, Azure PowerShell modules.

3

We can also deploy resources via the use of ARM templates and Bicep - based templates.

4

We can manage Azure - based resources from any client based machine - Linux, Windows or macOS.

Azure management and governance

1

With Azure resource manager templates, you can define your infrastructure as code. The template is in JSON format.

2

You can define the deployment of multiple resources in a template. You can maintain different versions of the template.

3

You can deploy the template using the Azure Portal or even with other tools such as Azure PowerShell.

4

Bicep templates can also be used to deploy resources by using a domain - specific language.

Azure management and governance

1

Azure Advisor tool - This is a cloud assistant that helps you to follow best practices when it comes to deployment of Azure resources.

2

Here you get recommendations that are divided into categories - Reliability, Security, Performance, Cost and Operational Excellence.

3

You can also generate alerts using Azure Advisor whenever there are new recommendations from Azure.

4

Azure Advisor has support for a variety of Azure - based services.

Azure management and governance

1

Azure Monitor tool - This is a monitoring solution that allows you to collect metrics and logs from Azure and on - premises.

2

You can define alert rules. These rules capture signals and based on conditions can send alerts as emails, SMS etc.

3

Application Insights which is a part of Azure Monitor offers the capability of application performance monitoring.

4

You can use a Log Analytics workspace to collect log data from your resources.

Azure management and governance

Which of the following can be used to collect any type of log data from Azure and non - Azure based resources?

Application Insights

Log Analytics workspace



Azure Policy

Azure management and governance

Which of the following page provides a global view of the health of Azure services and regions with their service availability?

Application Insights

Azure Advisor

Azure status



Azure management and governance

Which of the following can be used to enforce organizational standards and compliance of Azure - based resources.



Azure policies

Azure Advisor

Azure status

Azure management and governance

Your team has a set of Azure virtual machines. You have to ensure that a team of administrators are notified if an action of Deallocation is performed on the machines. Which of the following can you use for this requirement?

Azure policies

Azure Advisor

Azure Monitor

